

Tentative Budget – B.Tech Prototype

Sl. No.	Component / Activity	Estimated Cost (₹)
1	EM Tx/Rx coil fabrication and signal-conditioning components	2,500
2	Magnetometer + conductivity/electrical sensing components	3,500
3	Depth/pressure, temperature, and IMU sensors	2,500
4	ESP32/Arduino/Raspberry Pi-based data acquisition & control	3,500
5	Underwater camera + LED illumination	2,000
6	Small thrusters/motors, motor drivers, and power components	4,500
7	Prototype frame and waterproof electronics enclosure	3,500
8	Tether, cables, waterproof connectors, and miscellaneous electronics	2,500
9	Fabrication, calibration targets, and experimental consumables	2,000
10	Contingency/replacement components	1,500
	Estimated Total	₹28,000

Tentative Timeline – One Semester

Phase	Weeks	Major Activities	Expected Output
1. Literature Review & System Design	1–2	Study seabed mineral sensing; finalize sensors, ROV configuration, electronics, and experimental plan	System architecture & component selection
2. Sensor Development & Calibration	3–5	Develop EM Tx/Rx module; interface magnetometer, conductivity, depth, temperature, and IMU sensors; bench calibration	Working individual sensor modules
3. Prototype Fabrication & Integration	6–8	Fabricate frame/enclosure; integrate sensors, camera, thrusters, controller, and tether	Integrated VARUNA06 prototype

4. Data Acquisition & Processing	9–11	Develop data logging, filtering, calibration, and multi-sensor anomaly scoring/MPI algorithm	Processing and sensor-fusion framework
5. Controlled-Water Experiments	12–14	Test metallic/non-metallic targets at different positions/distances; repeat scans and evaluate responses	Experimental dataset & validated anomaly detection
6. Mapping & Final Demonstration	15–16	Demonstrate Detect → Score → Rescan → Confirm → Map, generate a small prospectivity/anomaly map, and document results	Final prototype, results, report & demonstration